

COLLEGE ALGEBRA AND TRIGONOMETRY



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College Algebra and Trigonometry

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CHAPTER OVERVIEW

1: Algebra Review

This College Algebra text will cover a combination of classical algebra and analytic geometry, with an introduction to the transcendental exponential and log arithmetic functions. There are also chapters devoted to the theory of functions, combinatorics and trigonometry. If mathematics is the language of science, then algebra is the grammar of that language. Like grammar, algebra provides a structure to mathematical notation, in addition to its uses in problem solving and its ability to change the appearance of an expression without changing the value.

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1.1: Algebraic Simplification

When algebraic techniques are presented as skills in isolation, they are much simpler to understand and practice. However the problem solving process in any context involves deciding which skills to use when. Most College Algebra students will have practiced problems in the form:

$$\begin{aligned}(x + 7)(x - 2) &=? \\ \text{or} \\ (2x + 1)^2 &=?\end{aligned}\tag{1.1.1}$$

The problems in this section deal with a combination of these processes which are often encountered as parts of more complex problems.

Example 1.1.1

Simplify:

$$3(x - 1)(2x + 5) - (x + 4)^2$$

Solution

In this example, the simplification involves two expressions: $3(x - 1)(2x + 5)$ and $(x + 4)^2$. The $(x + 4)^2$ is preceded by a negative (or subtraction) sign. This textbook will often treat $-x$ and $+(-x)$ as equivalent statements, since subtraction is defined as the addition of a negative.

We will simplify each expression separately and then look to combine like terms.

$$3(x - 1)(2x + 5) - (x + 4)^2 = 3(2x^2 + 3x - 5) - (x^2 + 8x + 16)$$

Notice that the results of both multiplications remain inside of parentheses. This is because each one has something that must be distributed.

In the case of $(2x^2 + 3x - 5)$, there is a 3 which must be distributed, resulting in $6x^2 + 9x - 15$. In the case of $(x^2 + 8x + 16)$ there is a negative sign or -1 which must be distributed, resulting in $-x^2 - 8x - 16$. It is important in these situations that the negative sign be distributed to all terms in the parentheses.

So

$$\begin{aligned}3(x - 1)(2x + 5) - (x + 4)^2 &= 3(2x^2 + 3x - 5) - (x^2 + 8x + 16) \\ &= 6x^2 + 9x - 15 - x^2 - 8x - 16 \\ &= 5x^2 + x - 31\end{aligned}\tag{1.1.2}$$

Example 1.1.2

Simplify:

$$2(x + 3)^2 - 4(3x - 1)(x + 2)$$

Solution

This example shows some of the same processes as the previous example. There are again two expressions that must be simplified, each of which has a coefficient that must be distributed. It is often helpful to wait until after multiplying the binomials before distributing the coefficient. However, as is often true in mathematics, there are several different approaches that may be taken in simplifying this problem.

If someone prefers to first distribute the coefficient before multiplying the binomials, then the coefficient must only be distributed to ONE of the binomials, but not both. For example, in multiplying $3 * 2 * 5 = 30$, we can first multiply $2 * 5 = 10$ and then $3 * 10 = 30$. Each factor is multiplied only once.

In the example above we can proceed as we did with the previous example:

$$\begin{aligned}
 2(x+3)^2 - 4(3x-1)(x+2) &= 2(x^2 + 6x + 9) - 4(3x^2 + 5x - 2) \\
 &= 2x^2 + 12x + 18 - 12x^2 - 20x + 8 \\
 &= -10x^2 - 8x + 26
 \end{aligned}
 \tag{1.1.3}$$

Or, we can choose to distribute the 4 first:

$$\begin{aligned}
 2(x+3)^2 - 4(3x-1)(x+2) &= 2(x^2 + 6x + 9) - (12x - 4)(x + 2) \\
 &= 2x^2 + 12x + 18 - (12x^2 + 20x - 8) \\
 &= 2x^2 + 12x + 18 - 12x^2 - 20x + 8 \\
 &= -10x^2 - 8x + 26
 \end{aligned}
 \tag{1.1.4}$$

Or, we can distribute the 4 as a negative. If we do this, then the sign in front of the parentheses will be positive:

$$\begin{aligned}
 2(x+3)^2 - 4(3x-1)(x+2) &= 2(x^2 + 6x + 9) + (-12x + 4)(x + 2) \\
 &= 2x^2 + 12x + 18 + (-12x^2 - 20x + 8) \\
 &= -10x^2 - 8x + 26
 \end{aligned}
 \tag{1.1.5}$$

Distributing the 2 in front of the squared binomial must also be handled carefully if you choose to do this. If you distribute the 2 before squaring the $(x+3)$, then the 2 will be squared as well. If you choose to distribute the 2, the $(x+3)^2$ must be written out as $(x+3)(x+3)$

$$\begin{aligned}
 2(x+3)^2 - 4(3x-1)(x+2) &= 2(x+3)(x+3) - 4(3x-1)(x+2) \\
 &= (2x+6)(x+3) - 4(3x^2 + 5x - 2) \\
 &= 2x^2 + 12x + 18 - 12x^2 - 20x + 8 \\
 &= -10x^2 - 8x + 26
 \end{aligned}
 \tag{1.1.6}$$

Most examples in this text will distribute the coefficients as the last step before combining like terms for a final answer.

Example 1.1.3

Simplify:

$$3x[5 - (2x + 7)] + (3x - 2)^2 - (x - 5)(x + 4)$$

This example has three expressions that should be simplified separately before combining like terms. In the first expression $3x[5 - (2x + 7)]$, we should simplify inside the brackets before distributing the $3x$

$$\begin{aligned}
 &3x[5 - (2x + 7)] + (3x - 2)^2 - (x - 5)(x + 4) \\
 &= 3x[5 - 2x - 7] + (3x - 2)^2 - (x - 5)(x + 4) \\
 &= 3x[-2x - 2] + (3x - 2)(3x - 2) - (x^2 - x - 20) \\
 &= -6x^2 - 6x + (9x^2 - 12x + 4) - x^2 + x + 20 \\
 &= -6x^2 - 6x + 9x^2 - 12x + 4 - x^2 + x + 20 \\
 &= 2x^2 - 17x + 24
 \end{aligned}
 \tag{1.1.7}$$

Exercises 1.1.1

Simplify each expression.

- $(x-2)[2x-2(3+x)] - (x+5)^2$
- $3x^2 - [7x - 2(2x-1)(3-x)]$
- $(a+b)^2 - (a+b)(a-b) - [a(2b-2) - (b^2-2a)]$
- $5x - 3(x-2)(x+7) + 3(x-2)^2$
- $(m+3)(m-1) - (m-2)^2 + 4$
- $(a-1)(a-2) - (a-2)(a-3) + (a-3)(a-4)$
- $2a^2 - 3(a+1)(a-2) - [7 - (a-1)]^2$
- $2(x-5)(3x+1) - (2x-1)^2$
- $6y + (3y+1)(y+2) - (y-3)(y-8)$

10. $6x - 4(x + 10)(x - 1) + (x + 1)^2$

Answers

1. $-x^2 - 16x - 13$
- 2.
3. $3b^2$
- 4.
5. $6m - 3$
- 6.
7. $-2a^2 + 19a - 58$
- 8.
9. $2y^2 + 24y - 22$
- 10.

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1.2: Factoring

This section will review three of the most common types of factoring - factoring out a Greatest Common Factor, Trinomial Factoring and factoring a Difference of Squares.

Greatest Common Factor

Factoring out a greatest common factor essentially undoes the distributive multiplication that often occurs in mathematical expressions. This factor may be monomial or polynomial, but in these examples, we will explore monomial common factors.

In multiplying $3xy^2(5x - 2y) = 15x^2y^2 - 6xy^3$ the monomial term $3xy^2$ is multiplied or distributed to both terms inside the parentheses. The process of **factorization** undoes this multiplication.

Example 1.2.1

Factor

$$7x^2 + 14x$$

Solution

This expression has two terms. The coefficients share a common factor of 7 and the only variable involved in this expression is x . The highest power of the variable that is shared by both terms is x^1 , so this is the power of x that can be factored out of both terms. The greatest common factor is $7x$

$$7x^2 + 14x = 7x(x + 2)$$

It isn't necessary to find the greatest common factor right away. In more complicated problems, the factoring can be accomplished in pieces, similar in fashion to reducing fractions.

Example 1.2.2

Factor

$$42x^2y^6 + 98xy^3 - 210x^3y^2$$

Solution

This expression has three terms. It's not immediately clear what the greatest common factor of the coefficients is, but they're all even numbers, so we could at least divide them all by 2. The $98xy^3$ term has an x^1 , which means that this is the highest power of x that we could factor out of all the terms. The $210x^3y^2$ has a y^2 , which is the highest power of y that can be factored out of all the terms. So we can at least proceed with these factors:

$$\begin{aligned} 42x^2y^6 + 98xy^3 - 210x^3y^2 &= 2xy^2 * 21xy^4 + 2xy^2 * 49y - 2xy^2 * 105x^2 \\ &= 2xy^2 (21xy^4 + 49y - 105x^2) \end{aligned} \quad (1.2.1)$$

Now, we didn't try very hard to find the greatest common factor in the beginning of this problem, so it's important that we continue to question whether or not there are any remaining common factors. The 21 and 49 clearly share a common factor of 7, so it would make sense to see if 105 is divisible by 7 as well. If we divide 105 by 7, we see that $105 = 7 * 15$. So, we can also factor out a common factor of 7 from the remaining terms in the parentheses.

$$\begin{aligned} 2xy^2 (21xy^4 + 49y - 105x^2) &= 2xy^2 (7 * 3xy^4 + 7 * 7y - 7 * 15x^2) \\ &= 7 * 2xy^2 (3xy^4 + 7y - 15x^2) \\ &= 14xy^2 (3xy^4 + 7y - 15x^2) \end{aligned} \quad (1.2.2)$$

Trinomial Factoring ($a = 1$)

Trinomial factoring undoes the multiplication of two binomials, and it comes in two flavors - simple and complex. The simplest form of trinomial factoring involves a trinomial expression in the form $ax^2 + bx + c$ in which the value of a is 1. This makes the task of factorization simpler than if the value of a is not 1.

Example 1.2.3

Factor $x^2 + 7x + 10$

Solution

In this example, the value of a is 1, which makes this type of trinomial factoring a little less difficult than it would otherwise be. Whether or not the value of a is 1 the fundamental issue that governs this type of factoring is the $+$ or $-$ sign of the constant term. In this problem, the constant term is positive. That means that we need to find factors of 10 that add up to 7. This is relatively straightforward:

$$x^2 + 7x + 10 = (x + 2)(x + 5)$$

A companion problem to this one is $x^2 - 7x + 10$. Notice that, in this case, the sign of the constant term is still positive, which means that we still need factors of 10 that add up to 7. This means we still need to use 2 and 5. However, in this case, instead of the $+10$ being produced from a multiplication of $(+2)(+5)$ it is the result of multiplying $(-2)(-5)$. This is what makes the 7 in the second example negative:

$$x^2 - 7x + 10 = (x - 2)(x - 5)$$

Example 1.2.4

Factor $x^2 + 3x - 10$

Solution

In this case, the sign of the constant term is negative. That means that we need to find factors of 10 that have a difference of 3. This is still 5 and 2.

$$x^2 + 3x - 10 = (x - 2)(x + 5)$$

The multiplication of the (-2) and the $(+5)$ produce the (-10) and the fact that the 2 and 5 have opposite signs creates the difference that gives us $(+3)$. A companion problem to this one is $x^2 - 3x - 10$. In this case, the sign of the constant term is still negative, which means that we still need factors of 10 that have a difference of 3. This means we still need to use 2 and 5. However in this case, instead of the $(+3)$ as the coefficient of the middle term, we'll need a (-3) . To do this we simply reverse the signs of the 2 and 5 from the previous problem:

$$x^2 - 3x - 10 = (x + 2)(x - 5)$$

Now the $(+2)(-5)$ gives us (-10) , but the $+2x - 5x$ gives us $(-3x)$ instead of $(+3x)$.

Example 1.2.5

Factor $x^2 + 11x - 42$

Solution

In this problem, the sign of the constant term is negative. That means that we need factors of 42 that have a difference of 11. A systematic exploration of all the factor pairs of 42 can help us to find the correct pair:

1	42
2	21
3	14
6	7

Here, we can see that the factors 3 and 14 have a difference of 11. This means that we will use these factors in our answer: $(x - 3)(x + 14)$

14). In determining how to place the + and - signs in the parentheses, we can refer back to the original problem: $x^2 + 11x - 42$. If we want a difference of $(+11x)$, then we'll need to have $a(+14)$ and $a(-3)$

$$x^2 + 11x - 42 = (x - 3)(x + 14)$$

Example 1.2.6

Factor $x^2 + 28x + 96$

Solution

In this problem, the sign of the constant term is positive. That means that we need factors of 96 that add up to 28. A systematic exploration of all the factor pairs of 96 can help us to find the correct pair:

1	96
2	48
3	32
4	24
6	16
8	12

Here, we can see that the factors 4 and 24 add up to 28. This means that we will use these factors in our answer: $(x + 4)(x + 24)$. In determining how to place the + and - signs in the parentheses, we can refer back to the original problem:

$x^2 + 28x + 96$. If we want 4 and 24 to add up to $(+28)$, then they should both be positive:

$$x^2 + 28x + 96 = (x + 4)(x + 24)$$

In building the charts of factor pairs in the previous two problems, nothing more difficult than dividing the constant term by the numbers 1, 2, 3, 4, 5, 6, ... and so on can help you to find the full list of factor pairs. If you don't get a whole number when dividing - for instance $96 \div 5 = 19.2$, then this number is not included in the list of factor pairs.

Trinomial Factoring ($a \neq 1$)

If the value of a is not 1, this means that, if the trinomial is factorable, at least one of its binomial factors also has a coefficient other than 1. For instance:

$$(2x + 7)(x - 3) = 2x^2 + 1x - 21$$

If we were to try undo this multiplication through the process of trinomial factoring, we should look to the sign of the constant term. In this example, the sign is negative. This still means that we will need to find factor pairs that produce a difference of $(+1x)$ as the middle term. However, in this scenario, it is not just the factors of 21 that are involved in producing the $(+1x)$, but the combination of the factors of 21 and the factors of the leading coefficient 2. The middle term $(+1x)$ comes from the multiplication of the $(2x)(-3)$ and the multiplication of $(+7)(+1x)$

$$(2x + 7)(x - 3)$$

$$\begin{aligned}(2x + 7)(x - 3) &= 2x^2 - 6x + 7x - 21 \\ &= 2x^2 + x - 21\end{aligned}\tag{1.2.3}$$

In trying to factor a trinomial like $2x^2 + x - 21$, we need to take this into consideration. For example, if we were to factor $3x^2 - 10x + 8$, we should first still look to the sign of the constant term, which, in this case, is positive. That means we want factor pairs that will add up to 10. But we have to take into consideration the interaction of the factors of the 3 with the factors of the 8. The 3 is a prime number, which means that we don't have a choice - it can only be split up into $3 * 1$, so we can start:

$$\text{Factor } 3x^2 + 10x + 8\tag{1.2.4}$$

$$(3x \quad ?)(x \quad ?)$$

Our options for filling in the question marks will come from the factors of 8, either $8 * 1$ or $4 * 2$. The process is by trial and error:

$$(3x \quad 8)(x \quad 1)$$

$$3x + 8x = 11x$$

$$(3x \quad 1)(x \quad 8)$$

$$24x + 1x = 25x$$

$$(3x \quad 4)(x \quad 2)$$

$$6x + 4x = 10x$$

$$(3x \quad 2)(x \quad 4)$$

$$12x + 2x = 14x$$

We can see that the choice above:

$$(3x \quad 4)(x \quad 2)$$

gives us the required $10x$ as the middle term. since the original problem was $3x^2 + 10x + 8$ we'll want to fill in the signs as both positive:

$$3x^2 + 10x + 8 = (3x + 4)(x + 2)\tag{1.2.5}$$

A second method for handling this type of factoring depends on how the factors of the leading coefficient and constant term interact with each other to produce the middle term. In this process, given the problem $3x^2 + 10x + 8$, we can multiply the first and last coefficient and then look at the factor pairs of the product:

$$3 * 8 = 24 \quad (1.2.6)$$

1	24
2	12
3	8
4	6

We can see that the factor pair of 24 that adds up to 10 is $6 * 4$. We proceed by splitting the $10x$ into $6x + 4x$ and then factor by grouping. If you are uncomfortable with factoring by grouping, then this is probably not a good method to try. However, if you are comfortable with factoring by grouping, the rest of the process is relatively straightforward:

$$3x^2 + 10x + 8 = 3x^2 + 6x + 4x + 8 \quad (1.2.7)$$

We then factor a common factor from the first two terms and the last two terms separately, and then factor out the common binomial factor of $(x + 2)$

$$\begin{aligned} 3x^2 + 10x + 8 &= 3x^2 + 6x + 4x + 8 \\ &= 3x(x + 2) + 4(x + 2) \\ &= (x + 2)(3x + 4) \end{aligned} \quad (1.2.8)$$

Example 1.2.7

Factor $7x^2 - 5x - 18$

Solution

In this example, the sign of the constant term is negative, which means that we'll need factor pairs that produce a difference of 5. The leading coefficient is 7, which is prime, so, again, the only way to split up the 7 is $7 * 1$

$$(7x \quad ?)(x \quad ?)$$

The options for filling in the question marks come from the factors of 18, for which there are three possibilities: $18 * 1$, $9 * 2$, or $6 * 3$. We'll try each of these factor pairs in place of the question marks:

$$\begin{array}{cc} (7x \quad 18)(x \quad 1) & (7x \quad 1)(x \quad 18) \\ \curvearrowright & \curvearrowright \\ 18x - 7x = 11x & 126x - 1x = 125x \end{array}$$

$$\begin{array}{cc} (7x \quad 9)(x \quad 2) & (7x \quad 2)(x \quad 9) \\ \curvearrowright & \curvearrowright \\ 14x - 9x = 5x & 63x - 2x = 61x \end{array}$$

$$(7x \quad 6)(x \quad 3)$$

$$21x - 6x = 15x$$

$$(7x \quad 3)(x \quad 6)$$

$$42x - 3x = 39x$$

The choice above:

$$(7x \quad 9)(x \quad 2)$$

gives us the required $5x$ as the middle term. since we're looking for a $(-5x)$, we'll make the 14 negative and the 9 positive: $-14x + 9x = -5x$

$$7x^2 - 5x - 18 = (7x + 9)(x - 2)$$

If we want to try the other method for factoring $7x^2 - 5x - 18$, we would multiply $7 * 18 = 126$, and then work to find factor pairs of 126 that have a difference of 5

1	126
2	63
3	42
6	21
7	18
9	14

Here, the last factor pair, $9 * 14$, has a difference of 5. So then we proceed to factor by grouping:

$$\begin{aligned}
 7x^2 - 5x - 18 &= 7x^2 + 9x - 14x - 18 \\
 &= x(7x + 9) - 2(7x + 9) \\
 &= (7x + 9)(x - 2)
 \end{aligned}
 \tag{1.2.9}$$

Notice that when the -2 was factored out from the last two terms $-14x - 18$, we ended up with $-2(7x + 9)$, because $(-2) * (+9)$

$= -18$. This is also important because in order to factor out the common binomial factor of $(7x + 9)$, this binomial must be exactly the same in both terms.

Difference of Squares

Factoring a difference of squares is actually a special form of trinomial factoring. If we consider a trinomial of the form $ax^2 + bx + c$, where c is a perfect square and negative, we will find something interesting about the possible values of b that make the trinomial factorable.

Example

$$\text{Consider } x^2 + bx - 36 \quad (1.2.10)$$

For this expression to be factorable, the middle coefficient b would need to be equal to the difference of any of the factor pairs of 36. If we look at the possible factor pairs, we see the following:

1	36
2	18
3	12
4	9
6	6

This means that the possible values for b that would make this expression factorable are:

$$\begin{aligned}
 36 - 1 &= 35 \rightarrow x^2 + 35x - 36 = (x + 36)(x - 1) \\
 18 - 2 &= 16 \rightarrow x^2 + 16x - 36 = (x + 18)(x - 2) \\
 12 - 3 &= 9 \rightarrow x^2 + 9x - 36 = (x + 12)(x - 3) \\
 9 - 4 &= 5 \rightarrow x^2 + 5x - 36 = (x + 9)(x - 4) \\
 6 - 6 &= 0 \rightarrow x^2 + 0x - 36 = x^2 - 36 = (x + 6)(x - 6)
 \end{aligned} \quad (1.2.11)$$

As we see, factoring $x^2 - 36$ means that the factors of the perfect square $36 = 6 * 6$ will cancel each other out leaving $0x$ in the middle. If there is a perfect square as the leading coefficient, then this number should be square rooted as well:

$$16x^2 - 25 = (4x + 5)(4x - 5) \quad (1.2.12)$$

In the example above, the $+20x$ and $-20x$ as the middle terms cancel each other out leaving just $16x^2 - 25$. These three types of factoring can also be combined with each other as we see in the following examples.

Example 1.2.6

Factor $2x^2 - 50$

Solution

This is not a trinomial because it doesn't have three terms. It is also not a difference of squares because 2 and 50 are not perfect squares. However, there is a common factor of 2 which we can factor out:

$$2x^2 - 50 = 2(x^2 - 25) \quad (1.2.13)$$

The expression inside the parentheses is a difference of squares and should be factored:

$$2x^2 - 50 = 2(x^2 - 25) = 2(x + 5)(x - 5) \quad (1.2.14)$$

Example 1.2.7

Factor $24 - 2x - x^2$

Solution

Here the sign of the x^2 term is negative. For this problem we can factor out -1 and proceed as we did with the previous problems in which the leading coefficient was positive or we can factor it as it is:

$$24 - 2x - x^2 = -(x^2 + 2x - 24) = -(x + 6)(x - 4) \quad (1.2.15)$$

If we want to factor it as it is, we should be aware that the constant term is positive and the quadratic term is negative, which means that we will want the factors of 24 to have a difference of 2

$$24 - 2x - x^2 = (6 + x)(4 - x) \quad (1.2.16)$$

Example 1.2.8

Factor $6x^2 + 12x + 6$

Solution

First, we notice that this expression has a common factor of 6. If we factor out the 6, then we should be left with an easier problem:

$$6x^2 + 12x + 6 = 6(x^2 + 2x + 1) = 6(x + 1)(x + 1) = 6(x + 1)^2 \quad (1.2.17)$$

Exercises 1.2.1

Factor each expression completely.

- 1) $8a^2b^3 + 24a^2b^2$
- 2) $19x^2y - 38x^2y^3$
- 3) $13t^8 + 26t^4 - 39t^2$
- 4) $5y^5 + 25y^4 - 20y^3$
- 5) $45m^4n^5 + 36mn^6 + 81m^2n^3$
- 6) $125x^3y^5 + 60x^4y^4 - 85x^5y^2$

Factor each trinomial into the product of two binomials.

- 7) $a^2 + 3a + 2$
- 8) $y^2 - 8y - 48$

- 9) $x^2 - 6x - 27$
- 10) $t^2 - 13t + 42$
- 11) $m^2 + 3m - 54$
- 12) $x^2 + 11x + 24$

Factor completely. Remember to look first for a common factor. If the polynomial is prime, state this.

- 13) $a^2 - 9$
- 14) $y^2 - 121$
- 15) $-49 + k^2$
- 16) $-64 + t^2$
- 17) $6x^2 - 54$
- 18) $25y^2 - 4$
- 19) $200 - 2a^2$
- 20) $3m^2 - 12$
- 21) $98 - 8k^2$
- 22) $-80w^2 + 45$
- 23) $5y^2 - 80$
- 24) $-4a^2 + 64$
- 25) $8y^2 - 98$
- 26) $24a^2 - 54$
- 27) $36k - 49k^3$
- 28) $16y - 81y^3$

Factor each trinomial completely. Remember to look first for a common factor. If the polynomial is prime, state this.

- 29) $3y^2 - 15y + 16$
- 30) $8a^2 - 14a + 3$
- 31) $9x^2 - 18x + 8$
- 32) $6a^2 - 17a + 12$
- 33) $2x^2 + 7x + 6$
- 34) $2m^2 + 13m - 18$
- 35) $20y^2 + 22y + 6$
- 36) $36x^2 + 81x + 45$
- 37) $24a^2 - 42a + 9$
- 38) $48x^2 - 74x - 10$

Factor each expression completely.

- 39) $30 + 7y - y^2$
- 40) $45 + 4a - a^2$
- 41) $24 - 10x - x^2$
- 42) $36 - 9x - x^2$
- 43) $84 - 8x - x^2$
- 44) $72 - 6a - a^2$
- 45) $6y^2 + 24y + 15$
- 46) $10y^2 - 75y + 35$
- 47) $20ax^2 - 36ax - 8a$

Answer

- 1) $8a^2b^2(b + 3)$
- 3) $13t^2(t^6 + 2t^2 - 3)$
- 5) $9mn^3(5m^3n^2 + 4n^3 + 9m)$
- 7) $(a + 2)(a + 1)$
- 9) $(x - 9)(x + 3)$
- 11) $(m + 9)(m - 6)$
- 13) $(a + 3)(a - 3)$

- 15) $(k + 7)(k - 7)$
- 17) $6(x + 3)(x - 3)$
- 19) $2(10 + a)(10 - a)$
- 21) $2(7 + 2k)(7 - 2k)$
- 23) $5(y + 4)(y - 4)$
- 25) $2(2y + 7)(2y - 7)$
- 27) $k(6 + 7k)(6 - 7k)$
- 29) **PRIME**
- 31) $(3x - 4)(3x - 2)$
- 33) $(2x + 3)(x + 2)$
- 35) $2(5y + 3)(2y + 1)$
- 37) $3(2a - 3)(4a - 1)$
- 39) $(10 - y)(3 + y)$
- 41) $(12 + x)(2 - x)$
- 43) $(14 + x)(6 - x)$
- 45) $3(2y^2 + 8y + 5)$
- 47) $4a(5x + 1)(x - 2)$

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1.3: Quadratic Equations

Quadratic equations are equations of the second degree. The solution of quadratic equations has a long history in mathematics going back several thousand years to the geometric solutions produced by the Babylonian culture. The Indian mathematician Brahmagupta used "rhetorical algebra" (algebra written out in words) in the 7 th century to produce solutions to quadratic equations and Arab mathematicians of 9 th and 10 th centuries followed similar methods. Leonardo of Pisa (also known as Fibonacci) included information on the Arab approach to solving quadratic equations in his book *Liber Abaci*, published in 1202

The quadratic formula is generally used to solve quadratic equations in standard form: $ax^2 + bx + c = 0$. The solutions for this are:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (1.3.1)$$

Now, the question is - why does this formula give solutions to the standard quadratic equation? We can proceed as we normally do in solving linear equations - that is, by getting the x by itself. The only problem here is that instead of just x , there are also terms involving x^2 . This is where the process of completing the square comes in handy.

We can begin with the quadratic equation in standard form:

$$ax^2 + bx + c = 0 \quad (1.3.2)$$

Just as it is easier to factor a quadratic trinomial if the leading coefficient is 1, this process of completing the square is also easier if the leading coefficient is 1. So next we will divide through on both sides of this equation by a .

$$\begin{aligned} \frac{ax^2 + bx + c}{a} &= \frac{0}{a} \\ \frac{ax^2}{a} + \frac{bx}{a} + \frac{c}{a} &= \frac{0}{a} \\ x^2 + \frac{b}{a}x + \frac{c}{a} &= 0 \end{aligned} \quad (1.3.3)$$

Then, we will move the $\frac{c}{a}$ to the other side of the equation to clear out some room for completing the square:

$$\begin{aligned} x^2 + \frac{b}{a}x + \frac{c}{a} &= 0 \\ -\frac{c}{a} &= -\frac{c}{a} \\ x^2 + \frac{b}{a}x &= -\frac{c}{a} \end{aligned} \quad (1.3.4)$$

Now we need to complete the square. If you are already familiar with this process, you may wish to skip the following explanation.

If we look at what happens when we square a binomial like $(x + 3)^2$, we will begin to notice a pattern.

$$\begin{aligned} (x + 3)^2 &= (x + 3)(x + 3) = x^2 + 6x + 9 \\ (x + 4)^2 &= (x + 4)(x + 4) = x^2 + 8x + 16 \\ (x + 5)^2 &= (x + 5)(x + 5) = x^2 + 10x + 25 \\ (x + 6)^2 &= (x + 6)(x + 6) = x^2 + 12x + 36 \end{aligned} \quad (1.3.5)$$

Our goal in the derivation of the Quadratic Formula is to rewrite the expression $x^2 + \frac{b}{a}x$ as a perfect square in the form $(x + \quad)^2$. The reason that we want to do this is that writing an expression as a binomial squared eliminates the problem of having both an x and an x^2 , which was preventing us from getting the x by itself in the standard quadratic equation.

If we can figure out what should take the place of the blanks in the statement:

$$x^2 + \frac{b}{a}x + \quad = (x + \quad)^2 \quad (1.3.6)$$

then we will be well on our way to deriving the quadratic formula.

If we re-examine the sample perfect binomial squares from the previous page, we note a useful pattern. This is that the blank in the parentheses $(x + \quad)^2$ is filled by a number that is one-half the value of the linear coefficient - or the coefficient of the x^1 term.

Notice that in $x^2 + 6x + 9 = (x + 3)^2$, 3 is half of 6, in $x^2 + 8x + 16 = (x + 4)^2$, the 4 is half of 8, and so on. If we want to write $x^2 + \frac{b}{a}x +$ as a perfect square in the form $(x + \quad)^2$, the blank in the parentheses should be filled by:

$$\frac{1}{2} * \frac{b}{a} = \frac{b}{2a} \quad (1.3.7)$$

Now, it's not true that $x^2 + \frac{b}{a}x + \quad = (x + \frac{b}{2a})^2$

We're missing the constant term on the left. However, if we return to our perfect square examples, we can see that the constant term is always the square of the term inside inside the parentheses. So, we can restate our problem now as:

$$\begin{aligned} x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 &= \left(x + \frac{b}{2a}\right)^2 \\ x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} &= \left(x + \frac{b}{2a}\right)^2 \end{aligned} \quad (1.3.8)$$

So, if we return to our original problem, we were saying that:

$$\begin{aligned} x^2 + \frac{b}{a}x + \frac{c}{a} &= 0 \\ -\frac{c}{a} &= -\frac{c}{a} \\ x^2 + \frac{b}{a}x &= -\frac{c}{a} \end{aligned} \quad (1.3.9)$$

We can add $\frac{b^2}{4a^2}$ to both sides of this equation and then restate the left hand side as a perfect square of a binomial:

$$\begin{aligned} x^2 + \frac{b}{a}x &= -\frac{c}{a} \\ + \frac{b^2}{4a^2} &= + \frac{b^2}{4a^2} \\ x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} &= -\frac{c}{a} + \frac{b^2}{4a^2} \\ \left(x + \frac{b}{2a}\right)^2 &= -\frac{c}{a} + \frac{b^2}{4a^2} \end{aligned} \quad (1.3.10)$$

The last tricky bit of this derivation is adding the two fractions on the right hand side. The common denominator for these fractions is $4a^2$, so we'll need to multiply the $-\frac{c}{a}$ by $\frac{4a}{4a}$ to get $-\frac{4ac}{4a^2}$. Then the right hand side will be $\frac{b^2 - 4ac}{4a^2}$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2} \quad (1.3.11)$$

Then, we can take the square root of both sides and get the x by itself:

$$\begin{aligned} \sqrt{\left(x + \frac{b}{2a}\right)^2} &= \pm \sqrt{\frac{b^2 - 4ac}{4a^2}} \\ x + \frac{b}{2a} &= \frac{\pm \sqrt{b^2 - 4ac}}{\sqrt{4a^2}} \\ x + \frac{b}{2a} &= \frac{\pm \sqrt{b^2 - 4ac}}{2a} \end{aligned} \quad (1.3.12)$$

Subtracting $\frac{b}{2a}$ from both sides is easy since we already have a common denominator:

$$\begin{aligned} x + \frac{b}{2a} &= \frac{\pm \sqrt{b^2 - 4ac}}{2a} \\ -\frac{b}{2a} &= -\frac{b}{2a} \\ x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \end{aligned} \quad (1.3.13)$$

At the College Algebra level, it is often useful to program the quadratic formula onto a graphing calculator both for easy use and also to learn a little bit about programming. The following program is a simple example of this for the TI84 series of graphing calculators. Graphing calculators also often have built-in polynomial solver feature that can be used to solve quadratics.

Press the "prgm" key in the top middle of the calculator keypad. This will bring up a screen that shows EXEC EDIT NEW across the top. Arrow over across the top to "NEW," and then select 1: Create New.

This will bring up a screen asking you to name the program. You should see PROGRAM and then underneath it, "Name=". The alpha lock is on automatically, so any key you press will type the letter associated with it. Name your program and press ENTER. You should see PROGRAM: Name, with whatever name you've chosen for your program. Underneath this you will see a colon: This is where you will enter the commands for the program.

First we need to enter the values for A , B and C from the quadratic equation into the calculator. To do this, press the "prgm" key again. Across the top of the screen you should see CTL I/O COLOR EXEC.

Arrow over to I/O. This is the "input/output" menu. Choose number 2:Prompt. This will return you to the program screen where you will see :Prompt under the name of the program. After :Prompt, type A, B, C . You'll need to use the "alpha" key to access the letters and the comma is right above the 7 key.

PROGRAM: Name (whatever name you've chosen should show here) :Prompt A, B, C

On the next line of the program, we will take the values of A, B and C and use them to calculate the values of the roots of the equation. Type in the following:

PROGRAM: Name :Prompt A, B, C

: $(-B + \sqrt{B^2 - 4AC}) / (2A) \rightarrow R$

: $(-B - \sqrt{B^2 - 4AC}) / (2A) \rightarrow S$

In typing these two lines it's important that when you type $-B$, you use the negative key next to the decimal point, rather than the subtraction key. The calculator is very picky about this. When you're typing the $B^2 - 4AC$, you'll need to use the subtraction key on the far right of the keypad.

Also notice the double parentheses - one set for the numerator of the fraction and one set for the square root. If you don't type this in correctly it will produce Wrong answers. The arrow in the formula stores the values of the answer in the variables R and S , and the arrow is produced by the "sto $\rightarrow$ " key just above the ON button in the lower left of the keyboard.

Now that we've given the calculator the values for A, B and C and then had the calculator find the roots of the equation, we need to display the answers. If you press the prgm key and arrow over to the I/O menu again, you can choose 3:Disp. This will Display the answers that we've stored as R and S .

PROGRAM: Name :Prompt $A, B, C : (-B + \sqrt{B^2 - 4AC}) / (2A) \rightarrow R$

: $(-B - \sqrt{B^2 - 4AC}) / (2A) \rightarrow S$

: Disp R, S

Now we can test the program with some simple equations. To run the program, press the program key and choose the program you've created either by selecting it and pressing enter, or by pressing the number for the program in the list. This should bring you back to the calculation screen, where you can run the program by pressing enter. The calculator should then ask you for the values of A, B and

C

Solve for $x : 2x^2 - x - 1 = 0$

In this example the values for A, B and C are:

$A = 2$

$B = -1$

$C = -1$

Again, it's important that you use the negative sign key next to the decimal point for the values of any negative coefficients and not the subtraction key. The calculator should return values of 1 and -0.5 as the solutions.

Solve for

$x : x^2 + x + 1 = 0$

In this example the values for A, B and C are:

$A = 1$

$B = 1$

$C = 1$

The calculator should return values of $-0.5 \pm 0.8660254038i$ as solutions.

If you get an error message saying "NONREAL AnswerS," you'll need to adjust the calculator setting to allow for complex valued answers. You can do this by pressing the "mode" key in the top left of the keypad and arrowing down to the line that reads "REAL a+bi re^{^(θi)}." You can then arrow over to "a+bi" and press enter. This will allow the calculator to compute complex valued

answers.

Something very important to remember about the quadratic formula is that the equation must be in standard form in order to identify the values of A, B and C to use in the formula. For example in the equation:

$$3x^2 - 7 = 2x \quad (1.3.14)$$

it is important to understand that the values of A, B and C come from the standard form of the equation and not the present form of the equation. There are several pitfalls to watch out for in this equation. First of all, the $2x$ is on the opposite side of the equation from the other terms. That means that the value of B IS NOT $+2$. Also, if we were to move the $2x$ to the other side to put the equation in standard form, it is not the order of the terms, but degree of the variable that determines whether a coefficient is identified as A, B or C

In moving the $2x$ to the other side of the equation, I have seen students put the term they're adding to that side as the last term. There is nothing wrong about this, but if you do that, you must be careful about identifying the values of A, B and C

$$\begin{aligned} 3x^2 - 7 &= 2x \\ -2x &= -2x \\ 3x^2 - 7 - 2x &= 0 \end{aligned} \quad (1.3.15)$$

There is nothing wrong about the way the equation above is written despite the fact that it is not in "standard form." The important thing to remember is that "A" is not the coefficient of whichever term is listed first. It is the coefficient of the quadratic, or x^2 term. Likewise, "B" is not the coefficient of the second term, but rather the coefficient of the linear, or x^1 term. And "C" is not whichever number comes last, but rather the value of the constant term. So in the equation above, however it is written, the value of A is $+3$, B is -2 and C is -7

Exercises 1.3.1

Solve for x in each equation. Round any irrational values to the nearest 1000 th.

- 1) $x^2 + 7x = 2$
- 2) $5x^2 - 3x = 4$
- 3) $\frac{3}{4}x^2 = \frac{7}{8}x + \frac{1}{2}$
- 4) $\frac{2}{3}x^2 - \frac{1}{3} = \frac{5}{9}x$
- 5) $2x^2 + (\sqrt{5})x - 3 = 0$
- 6) $3x^2 + x - \sqrt{2} = 0$
- 7) $2.58x^2 - 3.75x - 2.83 = 0$
- 8) $3.73x^2 + 9.74x + 2.34 = 0$
- 9) $5.3x^2 + 7.08x + 1.02 = 0$
- 10) $3.04x^2 + 1.35x + 1.234 = 0$
- 11) $7x(x + 2) + 5 = 3x(x + 1)$
- 12) $5x(x - 1) - 7 = 4x(x - 2)$
- 13) $14(x - 4) - (x + 2) = (x + 2)(x - 4)$
- 14) $11(x - 2) + (x - 5) = (x + 2)(x - 6)$

Answer

- 1) $x \approx 0.275, -7.275$
- 3) $x \approx 1.587, -0.420$
- 5) $x \approx 0.787, -1.905$
- 7) $x \approx -0.548, 2.002$
- 9) $x \approx -0.164, -1.172$
- 11) $x \approx -0.575, -2.175$
- 13) $x = 10, 5$

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1.4: Complex Numbers

Our number system can be subdivided in many different ways. The most basic form of mathematics is counting and almost all human cultures have words to represent numbers (the Pirahã of South America are a notable exception). Thus the most basic set of numbers is the set of counting numbers represented by the double barred $\mathbb{N}$: $\mathbb{N} = \{1, 2, 3, 4, 5, 6, 7, \dots\}$ (we will set aside the debate as to whether or not zero should be included in this set).

If we try to subtract a larger counting number from a smaller counting number we find that there are no members in the set of counting numbers to represent the answer in this situation. This extends the set of natural numbers to the set of integers: $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$. The integers are represented by the double barred $\mathbb{Z}$, for the German word for numbers-"zahlen." In the earliest appearances of negative numbers in the Chinese and Indian mathematical systems, negative values were often used to represent debt. Because Greek mathematics was based on Geometry, they did not use negative numbers.

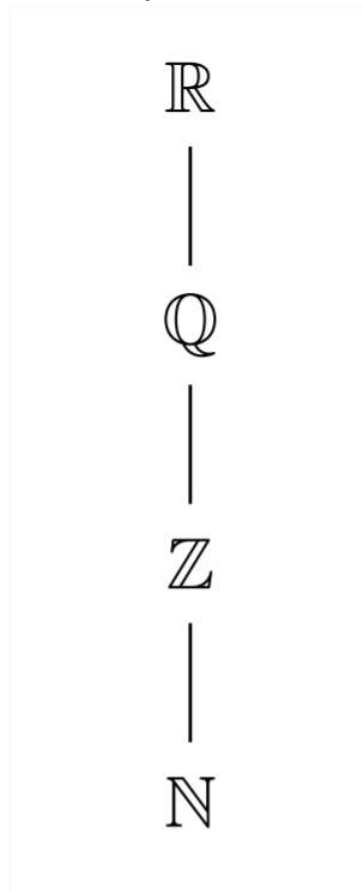
Moving to multiplication and division, if we question the value of $8 \div 2 = 4$ versus $8 \div 3 = ?$, we once again must expand our conception of numbers to allow for an answer to the second question $8 \div 3 = ?$. Understanding ratios of whole numbers or Rational numbers allows solutions to such problems. The set of Rational numbers is represented by the double barred $\mathbb{Q}$, to represent a quotient:

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z} \right\}$$

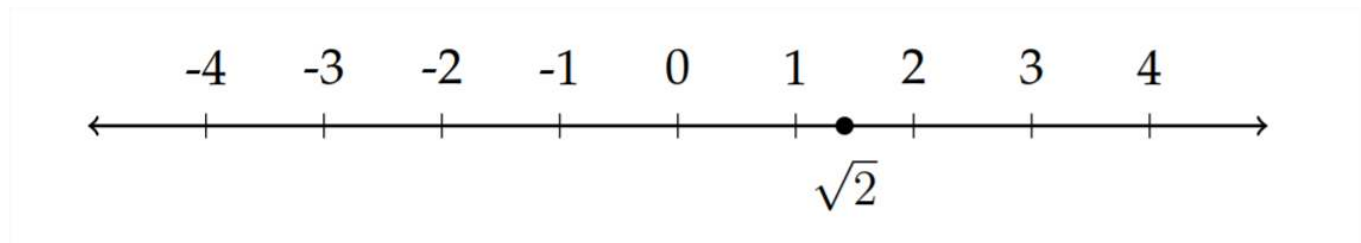
The Greek understanding of numbers mostly stopped here. They felt that all quantities could be represented as the ratio of whole numbers. The length of the diagonal of a square whose sides are of length 1 produced considerable consternation among the Pythagoreans as a result of this. Using the Pythagorean Theorem for the diagonal of a square whose sides are of length 1 shows that the diagonal would be $c^2 = 1^2 + 1^2 = 2$, thus $c = \sqrt{2}$. This number cannot be represented as a ratio of whole numbers. This new class of numbers adds the set

of irrational numbers to the existing set of rational numbers to create the Real numbers, represented with a double barred $\mathbb{R}$: $\mathbb{R}$.

This hierarchy of numbers is often represented in the following diagram:



One of the best ways to conceptualize the Real number system is on the number line - every point on the number line corresponds to a unique Real number and every Real number corresponds to a unique position on the Real number line.

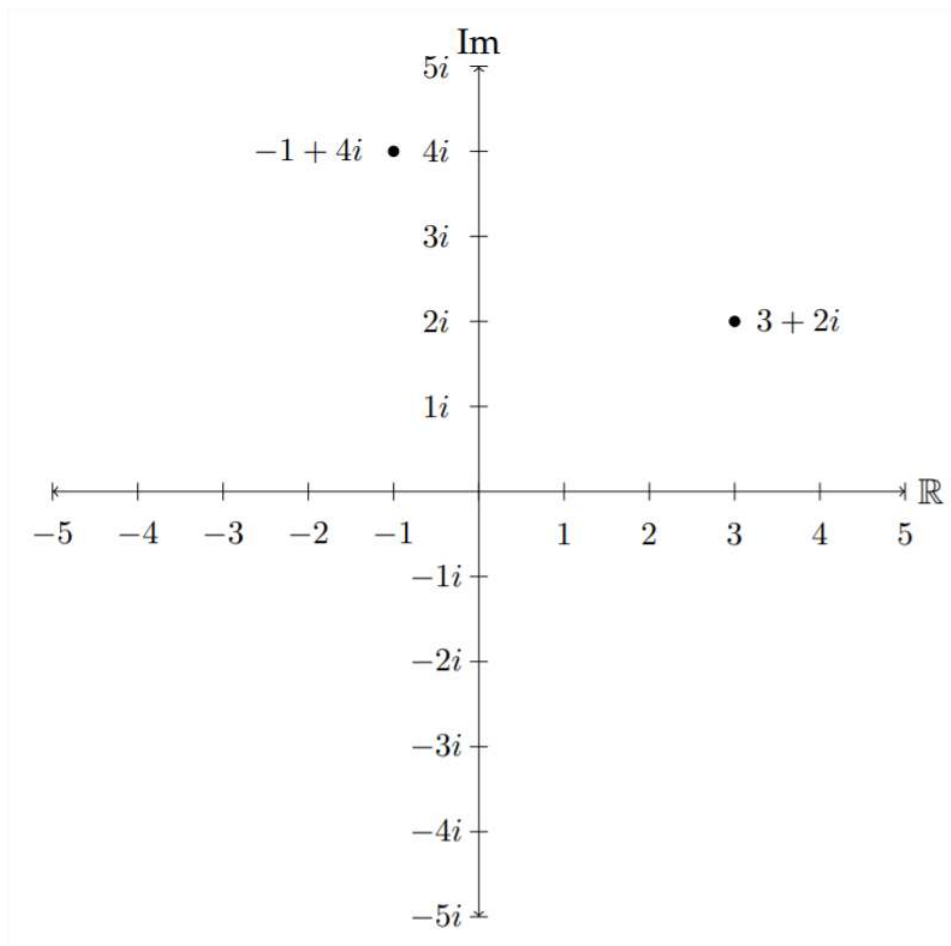


After the development of the printing press in the 15 th century, Fibonacci's Liber Abaci was translated into Italian from Latin and read throughout Italy. As a result, Italy became a thriving center of mathematics until the 17 th century, when the center of European mathematics moved north to France, Germany and England.

Throughout the 1500 's Italian mathematicians such as Girolamo Cardano, Raphael Bombelli and Niccolo Fontana Tartaglia worked to extend the ideas in Fibonacci's book. They produced formulas to solve cubic (x^3), and quartic (x^4) degree equations. In solving some of these equations they found that their formulas sometimes produced negative values under a square root. None of the known number systems could accomodate this possibility. In Cardano's book on algebra Ars Magna, he encounters a problem which involves the square root of a negative number. He says, "It is clear that this case is impossible. Nevertheless we will work thus..." and he proceeds to compute a valid complex solution to the problem. Mathematicians eventually defined the complex unit $\sqrt{-1} = i$ and then

devised a system in which all complex numbers are a combination of a Real part (a) and an "imaginary" part (bi).

The complex numbers are a two dimensional number system represented by the double barred $C : \mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$, where i is the complex unit defined as $i = \sqrt{-1}$. Throughout the late 1700 's and early 1800 's mathematicians gradually moved towards a geometrical interpretation of the two-dimensional complex numbers. What is today known as the "Argand diagram" represents the real valued portion of a complex number along a horizontal axis and the multiple of the complex unit along the vertical axis.



Expressing square roots of negative numbers

Square roots of negative quantities are generally expressed as a multiple of i

Example 1.4.1

$$\begin{aligned}\sqrt{-4} &= 2i \\ \sqrt{-25} &= 5i \\ \sqrt{-7} &\approx 2.646i\end{aligned}\tag{1.4.1}$$

Adding, subtracting and multiplying with complex numbers

Calculating with complex numbers has many similarities to working with variables. The real part and imaginary part are treated separately for addition and subtraction, but can be multiplied and divided.

Example 1.4.2

Compute the following:

$$\begin{aligned}(6 - 4i) + (-2 + 7i) &= 4 + 3i \\ (-9 + 2i) - (-4 + 6i) &= -9 + 2i + 4 - 6i = -5 - 4i \\ 3(10 + i) &= 30 + 3i \\ -7i(-5 + 8i) &= 35i - 56i^2\end{aligned}\tag{1.4.2}$$

Now we encounter an interesting fact about complex numbers and, in particular, the complex unit i . By definition, $i = \sqrt{-1}$. Therefore, if we square i we should get -1 . In the last example problem above, we can replace the i^2 with -1 to finish the problem.

$$\begin{aligned}
 -7i(-5 + 8i) &= 35i - 56i^2 \\
 &= 35i - 56(-1) \\
 &= 35i + 56 \\
 &= 56 + 35i
 \end{aligned}
 \tag{1.4.3}$$

Example 1.4.3

Compute the following:

$$\begin{aligned}
 (8 - 5i)(1 - 4i) &= 8 - 32i - 5i + 20i^2 \\
 &= 8 - 37i + 20(-1) \\
 &= 8 - 37i - 20 \\
 &= -12 - 37i
 \end{aligned}$$

$$\begin{aligned}
 (9 + 2i)^2 &= (9 + 2i)(9 + 2i) \\
 &= 81 + 18i + 18i + 4i^2 \\
 &= 81 + 36i + 4(-1) \\
 &= 81 + 36i - 4 \\
 &= 77 + 36i
 \end{aligned}
 \tag{1.4.4}$$

Powers of i

The powers of i follow an interesting pattern based on the definition that $i^2 = -1$

We can see that $i^1 = i$ and that $i^2 = -1$, as a result, $i^3 = i^2 * i^1 = -1 * i = -i$

In a similar fashion, $i^4 = i^2 * i^2 = (-1)(-1) = 1$

This means that $i^5 = i^4 * i = 1 * i = i$

If we put all of this information together we get the following:

$$\begin{aligned}
 i^1 &= i \\
 i^2 &= -1 \\
 i^3 &= -i \\
 i^4 &= 1 \\
 i^5 &= i^1 = i \\
 i^6 &= i^2 = -1 \\
 i^7 &= i^3 = -i \\
 i^8 &= i^4 = 1
 \end{aligned}$$

In other words, every power of i is equivalent to either i , -1 , $-i$, or 1 . To determine which of these values a power of i is equivalent to, we need to find the remainder of the exponent when it is divided by 4

Example 1.4.4

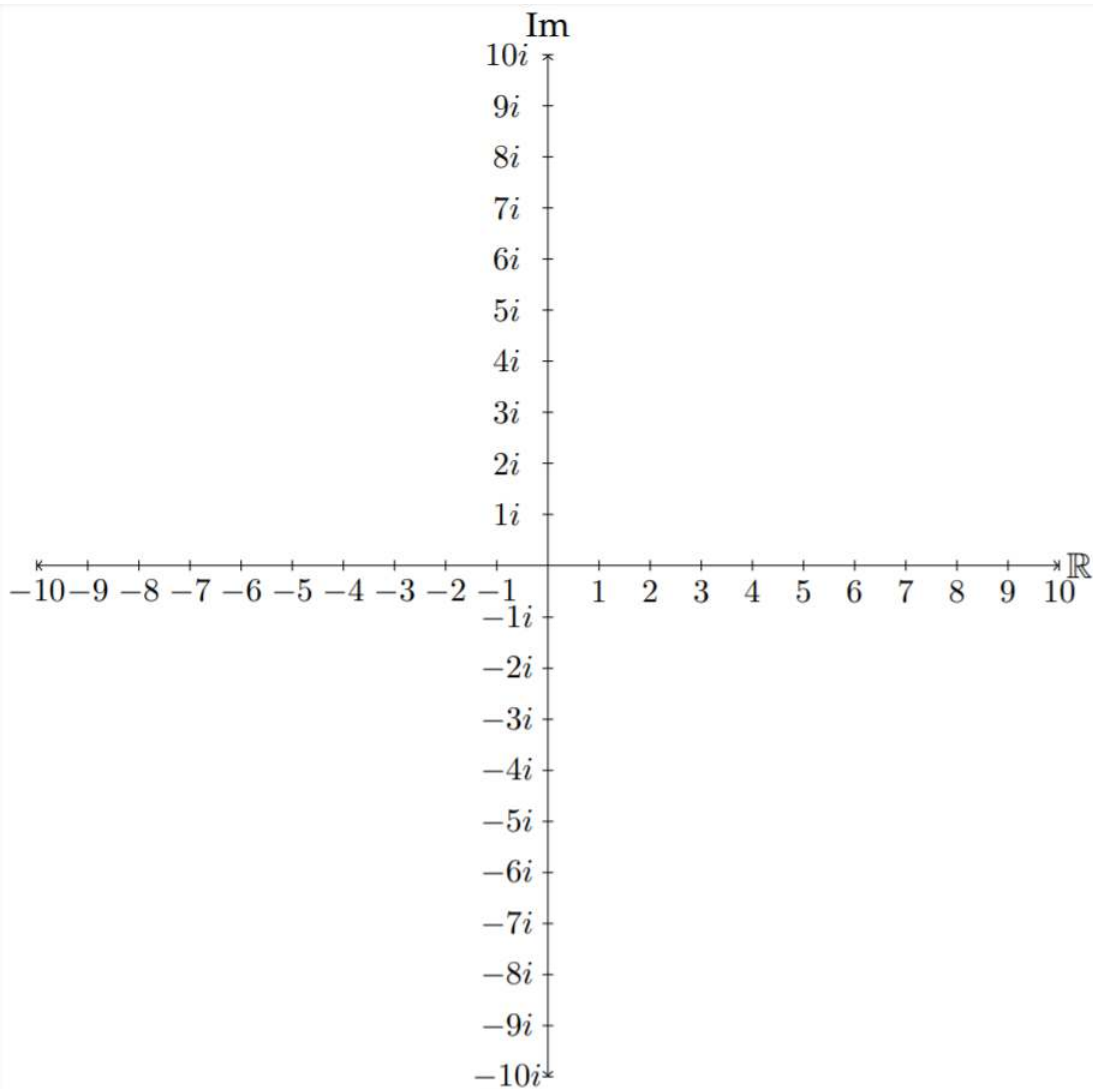
Simplify i^{38}

Solution

since every $i^4 = 1$, then $i^{38} = i^{36} * i^2 = (i^4)^9 * i^2 = 1^9 * i^2 = i^2 = -1$

since 38 is 2 more than a multiple of 4, then $i^{38} = i^2 = -1$

Exercise 1.4.1



Graph the following complex numbers:

- 1) $2 + 5i$
- 2) $4 - 3i$
- 3) $-2 + 6i$
- 4) $-3 - 5i$
- 5) 4
- 6) $-2i$
- 7) $7 - i$
- 8) $-1 + i$
- 9) $-8 + 4i$
- 10) $8 + 3i$
- 11) $7i$
- 12) $-5 - 9i$

Express each quantity in terms of i . Round irrational values to the nearest 1000 th.

- 13) $\sqrt{-36}$
- 14) $\sqrt{-81}$
- 15) $\sqrt{-100}$
- 16) $\sqrt{-49}$
- 17) $\sqrt{-4}$

- 18) $\sqrt{-25}$
- 19) $\sqrt{-2}$
- 20) $\sqrt{-6}$
- 21) $\sqrt{-10}$
- 22) $\sqrt{-31}$
- 23) $\sqrt{-5}$
- 24) $\sqrt{-3}$

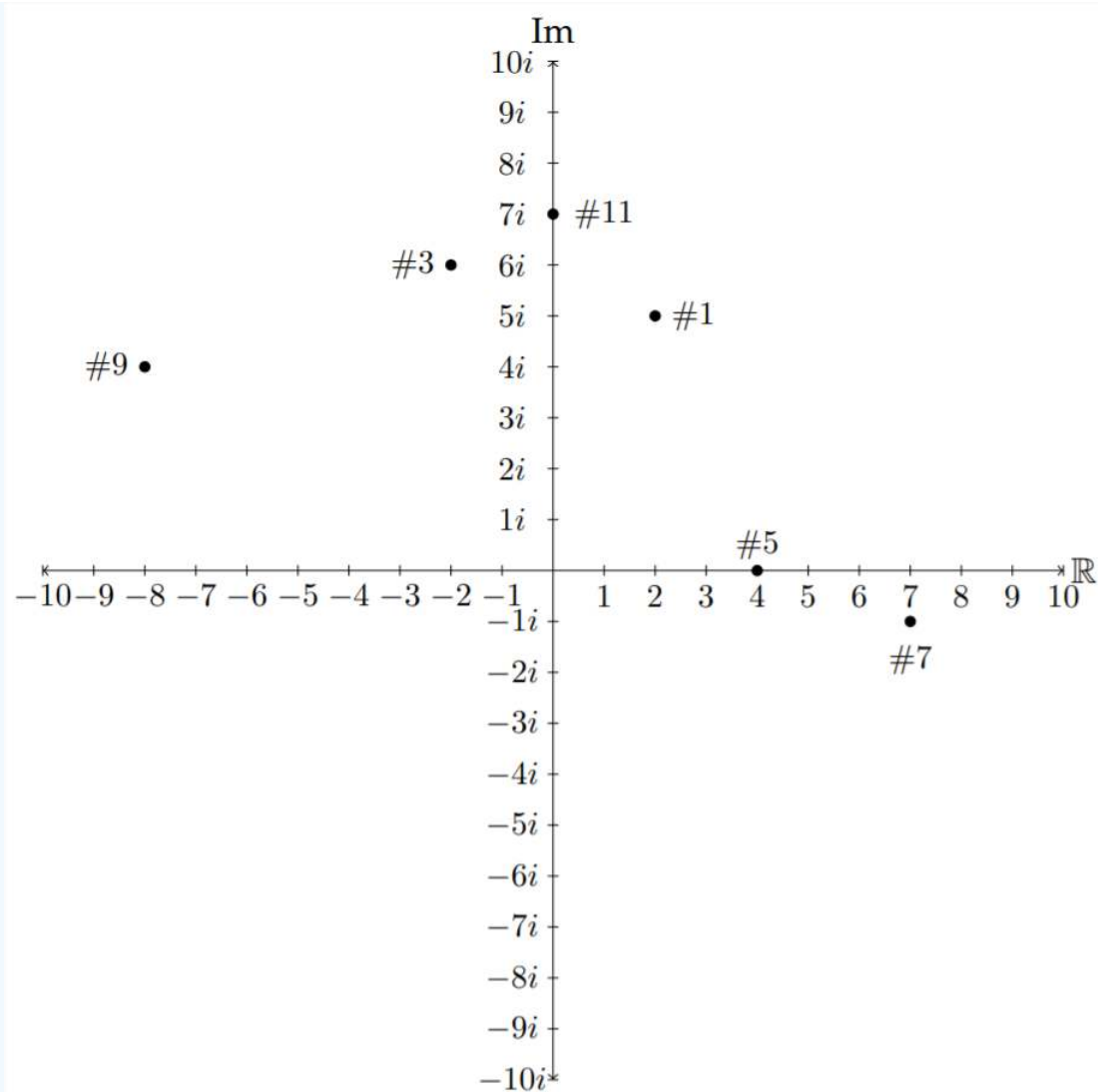
Perform the indicated operation and simplify

- 25) $(6 + 7i) + (5 + 3i)$
- 26) $(4 - 5i) + (3 + 9i)$
- 27) $(9 + 8i) - (1 - 2i)$
- 28) $(2 + i) - (6 - 4i)$
- 29) $(7 - 4i) - (5 - 3i)$
- 30) $(8 + i) - (4 + 3i)$
- 31) $(7i)(6i)$
- 32) $(4i)(-8i)$
- 33) $(-2i)(5i)$
- 34) $(12i)(3i)$
- 35) $(1 + i)(3 + 2i)$
- 36) $(1 + 5i)(4 + 3i)$
- 37) $(6 - 5i)(2 - 3i)$
- 38) $(8 - 3i)(2 + i)$
- 39) $(-3 + 4i)(-1 - 2i)$
- 40) $(-7 - i)(3 - 5i)$
- 41) $(4 - 2i)^2$
- 42) $(-5 + i)^2$
- 43) $(3 + i)(3 - i)$
- 44) $(2 + 6i)(2 - 6i)$
- 45) $(9 - 4i)(9 + 4i)$
- 46) $(5 + 2i)(5 - 2i)$

Express as either i , -1 , $-i$, or 1

- 47) i^3
- 48) i^7
- 49) i^{21}
- 50) i^{13}
- 51) i^{29}
- 52) i^{56}
- 53) i^{72}
- 54) i^{35}
- 55) i^{66}
- 56) i^{103}
- 57) i^{16}
- 58) i^{53}
- 59) i^{11}
- 60) i^{42}
- 61) i^{70}
- 62) i^9

Answer



- 13) $6i$
- 15) $10i$
- 17) $2i$
- 19) $1.414i$
- 21) $3.162i$
- 23) $2.236i$
- 25) $11 + 10i$
- 27) $8 + 10i$
- 29) $2 - i$
- 31) -42
- 33) 10
- 35) $1 + 5i$
- 37) $-3 - 28i$
- 39) $11 + 2i$
- 41) $12 - 16i$
- 43) 10
- 45) 97
- 47) $-i$
- 49) i
- 51) i

- 53) 1
- 55) -1
- 57) 1
- 59) $-i$
- 61) -1

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